
Online Functional Tensor-Train Decomposition via Continual Learning for Streaming Data Completion

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Abstract

Online tensor decompositions are powerful and proven techniques that address the challenges in processing high-dimensional tensor data, such as traffic flow and weather systems. The main aim of this work is to propose a more robust online tensor train decomposition (OTTD) framework, which represents a spatial-temporal continuous function using INRs continuity and easy adaptability for online data with the added compact and low representation of the Tensor-Train decomposition. The INRs allow for the natural characterization of continually expanding streaming data by simply adding new coordinates into the network. Particularly, our method transforms the classical online tensor decomposition algorithm into a more dynamic continual learning paradigm of updating the INR weights to fit the new data without forgetting the previous tensor knowledge in a compact manner. To this end, we introduce a long-tail memory replay method that adapts to the local continuity property of INR.

1 Introduction

Online tensor decomposition (2) is one of the foremost methods to address the streaming data completion problem by exploiting the potential compact structures underlying streaming data within an online optimization framework. However, decomposing tensor streams would lead to high computational costs owing to the significant growth in their volume over time. Also, dynamically capturing the internal latent properties of tensor streams, such as spatial-temporal continuity, poses difficult challenges. Novel research (1) (OFTD) created a new online learning paradigm that leverages INR's in order to learn a more dynamic and easy to expand model than the original decomposition parameters.

In this work, we propose an improvement over OFTD with a better, more robust model for streaming data completion. OTTD represents the spatial-temporal streaming data as a continuous function using TT tensor decomposition parameterized by implicit neural representations (INRs) (34; 35), which map an arbitrary spatial-temporal coordinate to the corresponding tensor value through deep neural networks. Instead of optimizing the factor matrices of tensor decomposition, OTTD optimizes the learnable weights of factor INRs during online optimization, which could better explore the complex dynamic structure and global spatial-temporal correlations of streaming data through a deep functional tensor representation. OTTD allows for natural characterizations of continually expanded streaming data by simply adding new coordinates into the INR network during online optimization. Although the tensor stream is continually growing, we maintain a constant number of optimization parameters by keeping the network structure unchanged.

Nevertheless, the incorporation of deep neural networks leads to the possible issue of forgetting the learned historical knowledge when fitting new data. Hence, to enable effective continual learning for processing streaming tensors using OTTD, we make a basic attempt by employing the memory

replay method, which utilizes a part of historical data during online optimization, to avoid forgetting. We develop a regret bound of OTTD from the perspective of the loss (i.e., forgetting) of historical knowledge, which theoretically shows that the OTTD model tends to forget the knowledge that is distant from the new tensor data due to the local continuity property of INRs. Hence we correspondingly design a memory buffer with a long-tail distribution (i.e., storing more data at more distant positions), which substantially enhances the performance for streaming data completion. Consequently, OTTD delivers superior performances against traditional online tensor decomposition methods, showcasing its strong ability for streaming data analysis.

We summarize the main contributions of this work as follows:

- We introduce a robust model for decomposing tensor data that is compact and is less sensitive to tensor rank.
- We find a new regret bound for our novel method in order to leverage memory replay method
- We apply OTTD to different data streams and compare it's predecessor OFTD

2 Related Work

2.1 Online Tensor Decomposition

Online analysis algorithms for streaming tensors have been widely developed in recent years (19; 20; 21). The CP decomposition is one of the mostly considered methods for streaming tensor completion. One issue with a CP decomposition is the sensitivity for the tensor rank (18) and finding the rank of a tensor is an NP-hard problem. These online tensor methods explicitly optimize factor matrices of the tensor decomposition using discrete online optimization rules. Different from these methods, the OFTD (1) employs INRs and CP decomposition to model streaming data as a spatial-temporal continuous function, which allows for natural characterization of continually expanded data by adding new coordinates into INRs during online optimization. Our proposed OTTD uses that method with a more robust tensor train model that should suffer less from the tensor rank.

2.2 Implicit Neural Representation

In recent years, INR has attracted widespread attention for their ability to continuously and implicitly represent signals through neural networks Sitzmann et al. (34), Mildenhall et al. (35). Compared to traditional discrete representations, INR offers advantages such as memory efficiency, continuous trajectory modeling, and analytical computations of higher-order derivatives. Typical applications of INR include 3D shape and scene reconstruction Atzmon and Lipman (36), Jiang et al. (37), view synthesis Mildenhall et al. (38), solving differential equations Chen et al. (39), and images/video processing Chen et al. (40), Skorokhodov et al. (41).

For instance, the well-known NeRF Mildenhall et al. (35) represents a scene as a continuous 5D function and optimizes neural radiance fields to synthesize novel views. Sitzmann et al. (34) introduced the SIREN, which leverages periodic activation functions to achieve high-fidelity representation of natural signals using INR. Chen et al. (40) proposed the local implicit image function, which encodes images as continuous functions and enables arbitrary-resolution image reconstruction. Currently, INR has become an increasingly popular research direction in mainstream AI fields Zhao et al. (42), Shi et al. (43), Li et al. (44). To our knowledge, the proposed method should be the first work to incorporate INR into the online tensor decomposition framework.

3 Proposed Method

3.1 Preliminaries

Definition 1 (Tensor Train Decomposition) . Given an N^{th} -order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, its TT decomposition is the representation using N factor tensors $\{U^{(n)}\}_{n=1}^N$ where the factor tensors $U^{(1)} \in \mathbb{R}^{I_1 \times r}$, $U^{(N)} \in \mathbb{R}^{r \times I_N}$ and $U^{(n)} \in \mathbb{R}^{r \times I_n \times r}$. Where r is some chosen parameter. We represent $\mathcal{X}$ with the following

$$\mathcal{X}_{i_1, \dots, i_N} = U_{i_1, :}^{(1)} \cdot U_{:, i_2, :}^{(2)} \cdot \dots \cdot U_{:, i_N}^{(N)}$$

And we denote $[[\{U^{(n)}\}_{n=1}^N]]$ as the N order tensor where each element is of the form above.

Definition 2 (Streaming tensor sequence). A sequence of N^{th} -order tensors $\{\mathcal{X}_t\}$ is called a streaming tensor sequence if for any $t \in \mathbb{Z}^+$, $\mathcal{X}_{t-1} \subseteq \mathcal{X}_t$. The t grows with time, and $\mathcal{X}_t$ is called the snapshot tensor taken at time t .

Definition 3 (Temporal tube). Given two successive tensors $\mathcal{X}_{t-1} \in \mathbb{R}^{I_1^{t-1} \times \dots \times I_N^{t-1}}$ and $\mathcal{X}_t \in \mathbb{R}^{I_1^t \times \dots \times I_N^t}$ derived from a streaming tensor sequence $\{\mathcal{X}_t\}$, the coming data (i.e., temporal tube) at time t can be represented by $\mathcal{Y}_t = \mathcal{X}_t \setminus \mathcal{X}_{t-1}$ which has the same size as $\mathcal{X}_t$, with entries given by:

$$(\mathcal{Y}_t)_{(i_1, \dots, i_N)} = \begin{cases} (\mathcal{X}_t)_{(i_1, \dots, i_N)} & \text{if } \exists n \text{ s.t. } I_n^{t-1} < i_n \leq I_n^t \\ 0 & \text{otherwise.} \end{cases}$$

3.2 Problem Formulation

Given a streaming tensor sequence $\{\mathcal{X}_t\}$ with missing entries, we aim to recover the missing data in the current snapshot $\mathcal{X}_t$. Since $\mathcal{X}_{t-1} \subseteq \mathcal{X}_t$, and we have handled $\mathcal{X}_{t-1}$ in previous time steps, the problem is equivalent to completing the elements in $\mathcal{Y}_t = \mathcal{X}_t \setminus \mathcal{X}_{t-1}$. The optimization problem at time t is typically formulated as:

$$\min_{\{U_t^{(n)}\}} \|\mathcal{P}_t \otimes (\mathcal{Y}_t - [[\{U_t^{(n)}\}_{n=1}^N]])\|_F \quad (1)$$

where $\{U_t^{(n)}\}$ denotes the set of tensor factors, $\mathcal{P}_t$ is a binary tensor representing missing and observed entries of $\mathcal{Y}_t$. Most tensor method rely on some regularization term, such as the total variation (46). Instead we use the spatial-temporal smoothness of data through the implicit smoothness of INRs (see Lemma 1).

3.3 Online Functional Tensor Decomposition

In this section, we give the detailed formulation of the proposed online functional tensor decomposition for streaming data completion.

Functional Tensor Decomposition Before introducing the online algorithm, we first introduce the batch tensor train decomposition (TTD) (i.e., off-line setting), which uses INRs to parameterize factor matrices of the TT decomposition. Specifically, for the n -th TT factor tensor $U_t^{(n)}$, we use an INR (34) to parameterize it. Such INR is a multilayer perceptron (MLP) where

with parameters Θ_n , which takes a coordinate $v \in \mathbb{R}$ as input and returns a vector or matrix accordingly:

$$f_{\Theta_n}(v) = W_d(\sigma(W_{d-1} \cdots \sigma(W_1 v))) \quad (2)$$

where $\Theta_n \triangleq \{W_i\}_{i=1}^d$ are weight matrices with $W_1 \in \mathbb{R}^{1 \times r}$ and $W_i \in \mathbb{R}^{r \times r}$ for $i = 2, 3, \dots, d-1$ and $W_d \in \mathbb{R}^{r \times r}$ or $W_d \in \mathbb{R}^{r \times r^2}$ depending on $n = 1, N$ or $n = 2, \dots, N-1$ and $\sigma(\cdot) = \sin(\omega_0 \cdot)$ is the sine activation function that is more effective for INR (34). To generate the factor tensors $U_t^{(n)}$, we consider the parallel notation for I_n^t input coordinates:

$$f_{\Theta_n}(v) \triangleq (f_{\Theta_n}(v_{(1)}), \dots, f_{\Theta_n}(v_{(I_n^t)}))^T \in \mathbb{R}^{I_n^t \times r},$$

where $v = [I_n^t] \triangleq (1, 2, \dots, I_n^t)^T$ represents the input coordinate vector and $v_{(i)}$ denotes the i -th element of the vector v . We use this output tensor $f_{\Theta_n}(v)$ (i.e., $f_{\Theta_n}([I_n^t])$) as the factor tensor $U_t^{(n)}$ of the TT decomposition and then obtain the TTD representation of an N^{th} -order streaming tensor $\mathcal{X}_t$:

$$\mathcal{X}_t = [[\{f_{\Theta_n}([I_n^t])\}_{n=1}^N]] \quad (3)$$

We have hence used N INRs $\{f_{\Theta_n}(\cdot)\}_{n=1}^N$ to parameterize the factor tensors of the TT decomposition.

The TTD enjoys two potential advantages for streaming data analysis. First, the functional representation naturally allows us to model streaming data in an efficient way by simply adding new coordinates into the model (i.e., by expanding the coordinates $[I_n^t]$) during online optimization. With the expanding of coordinates over time, the output of INR (i.e., the factor tensors) correspondingly increases in size to fit the larger size of the new data stream. Second, the TTD is less sensitive to the data rank and more robust general use model.

Online Functional Tensor Decomposition We now introduce the proposed OTTD. In the online setting, a new tensor $\mathcal{X}_t \in \mathbb{R}^{I_1^t \times I_2^t \times \dots \times I_N^t}$ arrives at each time t with missing values. We dynamically update the TTD model to accommodate the incremental growth of the streaming tensor. Our streaming tensor completion algorithm consists of the initialization stage and the online update stage.

Initialization Stage: Given an initial streaming tensor $\mathcal{X}_1 \in \mathbb{R}^{I_1^1 \times I_2^1 \times \dots \times I_N^1}$, we optimize the following objective function based on the TTD representation:

$$\min_{\{\Theta_n\}} \|\mathcal{P}_1 \otimes (\mathcal{Y}_1 - [\{f_{\Theta_n}([I_n^1])\}_{n=1}^N])\|_F^2, \quad (4)$$

where $\mathcal{P}_1$ is the initial mask, $\mathcal{Y}_1 = \mathcal{X}_1$ is the initial observed tensor, Θ_n are learnable parameters of the INR $f_{\Theta_n}(\cdot)$, and $[I_n^1]$ denotes the coordinates of the tensor along dimension n . To address the optimization problem, we use gradient descent-based methods to update the INR parameters $\{\Theta_n\}_{n=1}^N$ with the loss.

Online Stage: In the online update stage, a new tensor $\mathcal{X}_t \in \mathbb{R}^{I_1^t \times \dots \times I_N^t}$ ($t \geq 2$) comes at each time point and we need to update the online TTD model to accommodate the growth of streaming tensor sizes. We take the update of the n -th dimension as an example. At time t , the streaming tensor size grows by the scale of $I_n^t - I_n^{t-1}$ along the n -th dimension. We adapt to the new tensor size by simply adding $I_n^t - I_n^{t-1}$ coordinates to the corresponding factor INR $f_{\Theta_n}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^r$. Specifically, suppose that the coordinate vector at the last time point is $[I_n^{t-1}] \triangleq (1, \dots, I_n^{t-1})^T$, then the new coordinate vector at the time point t is $[I_n^t] \triangleq (1, \dots, I_n^t)^T$. Based on the enlarged OFTD model, we consider the following optimization problem during the online stage t :

$$\min_{\{\Theta_n\}} \|\mathcal{P}_t \otimes (\mathcal{Y}_t - [\{f_{\Theta_n}([I_n^t])\}_{n=1}^N])\|_F^2 \quad (5)$$

Similarly, we use the Adam optimizer to update the INR parameters $\{\Theta_n\}_{n=1}^N$ with the updated loss at each time point t . Here, the observed data $\mathcal{Y}_t = (\mathcal{X}_t \setminus \mathcal{X}_{t-1}) \in \mathbb{R}^{I_1^t \times \dots \times I_N^t}$ contains the new data at time t , i.e., the INRs $\{f_{\Theta_n}(\cdot)\}_{n=1}^N$ are optimized to fit the new data in $\mathcal{X}_t \setminus \mathcal{X}_{t-1}$ with initialization weights $\{\Theta_n\}$ being taken from the last time point (i.e., the continual learning). Such online optimization strategy is reasonable since it is impractical to use all historical data at each time point, as this would result in large computational costs. Therefore, we consider only fitting the new data to optimize the INRs in each time point t . For single-aspect streams, we continually expand the temporal coordinates of OTTD, while for multi-aspect streams, we need to continually expand the coordinates in all dimensions $n = 1, \dots, N$. Given a total time T , the completion result is obtained by the FTD model $[\{f_{\Theta_n}([I_n^T])\}_{n=1}^N] \in \mathbb{R}^{I_1^T \times \dots \times I_N^T}$ after T optimization steps under the continual learning manner.

3.4 Theoretical Analysis

We interpret two insights of our method, i.e., the spatial-temporal continuity of OTTD brought from INR, and the regret bound of the OTTD model that reveals its forgetting behavior, which motivates us to design memory replay to alleviate forgetting. First, we show that our OTTD method preserves the tensor spatial-temporal smoothness from the Lipschitz smooth perspective. Lemma 1 shows that the TTD model preserves the spatial-temporal continuity of the estimated tensor $\mathcal{X}_t$ (i.e., elements that are closer to each other are more likely to share similar structures).

Based on Lemma 1, we present a regret bound of OTTD regarding the loss of historical information when fitting new data streams, which reveals the forgetting behavior of INR.

Theorem 1 shows that the regret bound is proportional to the positional distance $(I_n^t + 1 - i_n)$ between the considered point (i_1, i_2, i_3) and the new data stream position $I_n^t + 1$. This indicates that information that is more distant from the new data stream is more likely to be forgotten. To alleviate the forgetting, we design a long-tail memory replay continual learning method. [Lipschitz

smooth bound for FTD] Let the tensor $\mathcal{X}_t \in \mathbb{R}^{I_1^t \times I_2^t \times I_3^t}$ satisfy FTD, where each factor function is of the form $f_{\Theta_n}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^r$ ($n = 1, 3$) and $f_{\Theta_2}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^{r^2}$ where both are INR's formulated as $f_{\Theta_n}(v) = W_d(\sigma(W_{d-1} \cdots \sigma(W_1 v)))$ with activation function $\sigma(\cdot) = \sin(\omega_0 \cdot)$. Assume that each element of the weight matrix W follows i.i.d. $\mathcal{N}(0, w^2)$. Then for any $\delta \in (0, 1)$, any spatial coordinates $v, v' \in \mathbb{Z}^2$, where $v = (v_{(1)}, v_{(2)})$, $v' = (v'_{(1)}, v'_{(2)})$ and any temporal coordinates $k, k' \in \mathbb{Z}$, with probability at least $1 - \delta$ the following Lipschitz smoothness holds for $\mathcal{X}_t$:

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v'_{(1)}, v'_{(2)}, k)}| \leq C_1 \|v - v'\|_{l_1}, \quad (\text{Spatial smooth}) \quad (6)$$

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k')}| \leq C_1 |k - k'|, \quad (\text{Temporal smooth}) \quad (7)$$

where $C_1 = \omega_0^{3d-3} (2wr^3 + w \ln \frac{3d}{\delta})^{3d} \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$ is a Lipschitz constant.

Since each element of the weight matrices $W \in \mathbb{R}^{r \times r}$ follows i.i.d. $\mathcal{N}(0, w^2)$, we have the tail bound

$$\mathbb{P}(\|W\|_{l_1} \leq \eta) \geq 1 - \exp(-\eta/w + 2r^2) \quad (8)$$

We denote d to be the number of layers in and INR so we have $2d$ layers from $f_{\Theta_1}, f_{\Theta_3}$ with each layer $W \in \mathbb{R}^{r \times r}$ and another $d - 1$ layers with $W \in \mathbb{R}^{r \times r}$ and a single layer with $W \in \mathbb{R}^{r \times r^2}$. Hence from the previous lemma the probability of all the weight matrices of the OFTD model have l_1 -norm less than η is at least

$$1 - (3d - 1) \exp(-\eta/w + 2r^2) - \exp(-\eta/w + 2r^3) \geq 1 - 3d \exp(-\eta/w + 2r^3)$$

Let $1 - 3d \exp(-\eta/w + 2r^3) = 1 - \delta$, we have $\eta = 2wr^3 + w \ln \frac{3d}{\delta}$.

Note that the sine activation function satisfies $|\sin(\omega_0 x)| \leq \omega_0 |x|$ ($\omega_0 > 0$). Hence we have $\|f_{\Theta_n}(x)\|_{l_1} \leq \eta^d \omega_0^{d-1} |x|$. For any (i, j, k) we have

$$(\mathcal{X}_t)_{(i,j,k)} = f_{\Theta_1}([I_1^t]_{(i)}^T) f_{\Theta_2}([I_2^t]_{(j)}) f_{\Theta_3}([I_3^t]_{(k)}). \quad (9)$$

Where $f_{\Theta_1}([I_1^t]_{(i)}) \in \mathbb{R}^r$ and $f_{\Theta_3}([I_3^t]_{(k)}) \in \mathbb{R}^r$ and $f_{\Theta_2}([I_2^t]_{(j)}) \in \mathbb{R}^{r \times r}$. We first note that $f_{\Theta_2}([I_2^t]_{(j)}) \in \mathbb{R}^{r \times r}$ is a matrix and we can get the following loose upper bound:

$$\|f_{\Theta_2}([I_2^t]_{(j)})\|_{op} \leq \|f_{\Theta_2}([I_2^t]_{(j)})\|_{l_1}$$

We will use this loose upper bound in our proof to match our union bound we derived. Then for any (i, j, k) and (i', j, k) we have

$$\begin{aligned} |(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i',j,k)}| &= \left| (f_{\Theta_1}([I_1^t]_{(i)}^T) - f_{\Theta_1}([I_1^t]_{(i')}^T)) f_{\Theta_2}([I_2^t]_{(j)}) f_{\Theta_3}([I_3^t]_{(k)}) \right| \\ &\leq \|f_{\Theta_1}([I_1^t]_{(i)}^T) - f_{\Theta_1}([I_1^t]_{(i')}^T)\|_{l_1} \|f_{\Theta_2}([I_2^t]_{(j)})\|_{l_1} \|f_{\Theta_3}([I_3^t]_{(k)})\|_{l_1} \\ &= \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} \|f_{\Theta_2}(j)\|_{l_1} \|f_{\Theta_3}(k)\|_{l_1} \end{aligned} \quad (10)$$

The bound on the first term holds as:

$$\begin{aligned} \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} &= \|W_d(\sigma(W_{d-1} \cdots \sigma(W_1 i))) - W_d(\sigma(W_{d-1} \cdots \sigma(W_1 i')))\|_{l_1} \\ &\leq \eta \|\sigma(W_{d-1} \cdots \sigma(W_1 i)) - \sigma(W_{d-1} \cdots \sigma(W_1 i'))\|_{l_1} \\ &\leq \eta \omega_0 \|W_{d-1} \cdots \sigma(W_1 i) - W_{d-1} \cdots \sigma(W_1 i')\|_{l_1} \\ &\leq \eta^d \omega_0^{d-1} |i - i'|. \end{aligned} \quad (11)$$

Thus we have

$$\begin{aligned} |(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i',j,k)}| &\leq \|f_{\Theta_1}(i) - f_{\Theta_1}(i')\|_{l_1} \|f_{\Theta_2}(j)\|_{l_1} \|f_{\Theta_3}(k)\|_{l_1} \\ &\leq \eta^d \omega_0^{d-1} |i - i'| \eta^d \omega_0^{d-1} |j| \eta^d \omega_0^{d-1} |k| \\ &\leq \eta^{3d} \omega_0^{3d-3} I_2^t I_3^t |i - i'|. \end{aligned} \quad (12)$$

Similarly, we can derive the analogous bounds for the two other dimensions:

$$|(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i,j',k)}| \leq \eta^{3d} \omega_0^{3d-3} I_1^t I_3^t |j - j'| \quad (13)$$

$$|(\mathcal{X}_t)_{(i,j,k)} - (\mathcal{X}_t)_{(i,j,k')}| \leq \eta^{3d} \omega_0^{3d-3} I_1^t I_2^t |k - k'| \quad (14)$$

Hence from the triangle inequality one can easily derive that for any spatial coordinates v, v' and temporal coordinates k, k' the following bounds hold with probability at least $1 - \delta$:

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v'_{(1)}, v'_{(2)}, k)}| \leq C_1 \|v - v'\|_{l_1}, \quad (15)$$

$$|(\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k)} - (\mathcal{X}_t)_{(v_{(1)}, v_{(2)}, k')}| \leq C_1 |k - k'|, \quad (16)$$

where $C_1 = \omega_0^{3d-3}(2wr^3 + w \ln \frac{3d}{\delta})^{3d} \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$. The proof is completed.

[Regret bound for online FTD] Denote the OFTD model learned at the historical time point t by $\mathcal{X}_t = [[\{f_{\Theta_n}([I_n^t]; t)\}_{n=1}^3]] \in \mathbb{R}^{I_1^t \times I_2^t \times I_3^t}$, where $f_{\Theta_n}([I_n^t]; t)$ denotes the n -th factor function at time t . Assume that:

- The OFTD model learned at the new time point $t+1$ is invariant at the boundary (I_1^t, I_2^t, I_3^t) , i.e., $f_{\Theta_n}(I_n^t; t+1) = f_{\Theta_n}(I_n^t; t)$ ($n = 1, 2, 3$).
- Each element of the weight matrix of the d -layer INRs $\{f_{\Theta_n}(\cdot)\}$ follows i.i.d. $\mathcal{N}(0, w^2)$ with sine activation function $\sigma(\cdot) = \sin(\omega_0 \cdot)$.
- The l_1 -norm of derivative of each factor INR $\|f'_{\Theta_n}(x)\|_{l_1}$ is bounded by $\kappa > 0$.

Then for any $\delta \in (0, 1)$ and any historical position (i_1, i_2, i_3) ($i_n \leq I_n^t$), the following regret bound between the new OFTD model $\mathcal{X}_{t+1} = [[\{f_{\Theta_n}([I_n^{t+1}]; t+1)\}_{n=1}^3]] \in \mathbb{R}^{I_1^{t+1} \times I_2^{t+1} \times I_3^{t+1}}$ (Take an example $I_n^{t+1} = I_n^t + 1$) and the old OFTD model $\mathcal{X}_t$ holds with probability at least $1 - \delta$:

$$|(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \leq C_2 \max_n(I_n^t + 1 - i_n), \quad (17)$$

where $C_2 = 6(\eta^{3d} \omega_0^{3d-3} + \kappa \eta^{2d} \omega_0^{2d-2}) \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$ and $\eta = 2wr^3 + w \ln \frac{3d}{\delta}$.

First we derive the regret bound for each factor INR $f_{\Theta_n}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^r$. For any $x \in [0, I_n^t]$ we have:

$$\begin{aligned} \|f_{\Theta_n}(x; t+1) - f_{\Theta_n}(x; t)\|_{l_1} &\leq \left\| f_{\Theta_n}(x; t+1) - \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t+1) ds \right\|_{l_1} \\ &\quad + \left\| \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t+1) ds - \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t) ds \right\|_{l_1} \\ &\quad + \left\| \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t) ds - f_{\Theta_n}(x; t) \right\|_{l_1}. \end{aligned} \quad (18)$$

From Lemma 1 and its proof, we have that with probability at least $1 - \delta$ the first term on the right-hand side of (18) admits:

$$\begin{aligned} \left\| f_{\Theta_n}(x; t+1) - \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t+1) ds \right\|_{l_1} &= \left\| \int_{I_n^t}^{I_n^t+1} (f_{\Theta_n}(x; t+1) - f_{\Theta_n}(s; t+1)) ds \right\|_{l_1} \\ &\leq \left\| \int_{I_n^t}^{I_n^t+1} \eta^d \omega_0^{d-1} (s - x) ds \right\|_{l_1} \\ &\leq \left\| \int_{I_n^t}^{I_n^t+1} \eta^d \omega_0^{d-1} (I_n^t + 1 - x) ds \right\|_{l_1} \\ &= \eta^d \omega_0^{d-1} |I_n^t + 1 - x|, \end{aligned} \quad (19)$$

where $\eta = 2wr^2 + w \ln \frac{3d}{\delta}$. Similarly we can get for the third term:

$$\left\| \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t) ds - f_{\Theta_n}(x; t) \right\|_{l_1} \leq \eta^d \omega_0^{d-1} |I_n^t + 1 - x| \quad (20)$$

Define $h_n(x) = f_{\Theta_n}(x; t+1) - f_{\Theta_n}(x; t)$, and from assumptions we have $h_n(I_n^t) = 0$. From the mean value theorem for integrals, the second term on the right-hand side of (18) admits:

$$\left\| \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t+1) ds - \int_{I_n^t}^{I_n^t+1} f_{\Theta_n}(s; t) ds \right\|_{l_1} = \left\| \int_{I_n^t}^{I_n^t+1} h_n(s) ds \right\|_{l_1} = \|h_n(\zeta)\|_{l_1} \quad (21)$$

where $\zeta \in (I_n^t, I_n^t + 1)$. By the Lagrange mean value theorem we have:

$$\|h_n(\zeta)\|_{l_1} = \|h_n(\zeta) - h_n(I_n^t)\|_{l_1} = \|h'_n(\xi)(\zeta - I_n^t)\|_{l_1} \leq 2\kappa|I_n^t + 1 - x| \quad (22)$$

where $\xi \in (I_n^t, \zeta)$. Here, we have used the assumption that $\|h'_n(\xi)\|_{l_1} = \|f'_{\Theta_n}(\xi; t + 1) - f'_{\Theta_n}(\xi; t)\|_{l_1} \leq 2\kappa$. Combining all the results, we obtain the regret bound for the individual factor INRs:

$$\|f_{\Theta_n}(x; t + 1) - f_{\Theta_n}(x; t)\|_{l_1} \leq 2(\eta^d \omega_0^{d-1} + \kappa)|I_n^t + 1 - x|. \quad (23)$$

Now we can derive the regret bound for online FTD. Write each element of the tensor $\mathcal{X}_t$ as

$$(\mathcal{X}_t)_{(i_1, i_2, i_3)} = f_{\Theta_1}([I_1^t]; t)_{(i_1)} f_{\Theta_2}([I_2^t]; t)_{(i_2)} f_{\Theta_3}([I_3^t]; t)_{(i_3)}$$

By adding and subtracting terms (standard product difference expansion), for any (i_1, i_2, i_3) ($i_n \leq I_n^t$) we have:

$$\begin{aligned} & |(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \\ &= |f_{\Theta_1}([I_1^t]; t + 1)_{(i_1)}^T f_{\Theta_2}([I_2^t]; t + 1)_{(i_2)} f_{\Theta_3}([I_3^t]; t + 1)_{(i_3)} - f_{\Theta_1}([I_1^t]; t)_{(i_1)}^T f_{\Theta_2}([I_2^t]; t)_{(i_2)} f_{\Theta_3}([I_3^t]; t)_{(i_3)}| \\ &= |(f_{\Theta_1}(i_1; t + 1) - f_{\Theta_1}(i_1; t))^T f_{\Theta_2}(i_2; t) f_{\Theta_3}(i_3; t) \\ &\quad + f_{\Theta_1}(i_1; t + 1)^T (f_{\Theta_2}(i_2; t + 1) - f_{\Theta_2}(i_2; t)) f_{\Theta_3}(i_3; t) \\ &\quad + f_{\Theta_1}(i_1; t + 1)^T f_{\Theta_2}(i_2; t + 1) (f_{\Theta_3}(i_3; t + 1) - f_{\Theta_3}(i_3; t))| \\ &\leq \|f_{\Theta_1}(i_1; t + 1) - f_{\Theta_1}(i_1; t)\|_{l_1} \|f_{\Theta_2}(i_2; t)\|_{l_1} \|f_{\Theta_3}(i_3; t)\|_{l_1} \\ &\quad + \|f_{\Theta_2}(i_2; t + 1) - f_{\Theta_2}(i_2; t)\|_{l_1} \|f_{\Theta_1}(i_1; t + 1)\|_{l_1} \|f_{\Theta_3}(i_3; t)\|_{l_1} \\ &\quad + \|f_{\Theta_3}(i_3; t + 1) - f_{\Theta_3}(i_3; t)\|_{l_1} \|f_{\Theta_1}(i_1; t + 1)\|_{l_1} \|f_{\Theta_2}(i_2; t + 1)\|_{l_1}. \end{aligned} \quad (24)$$

Substituting (23) and the bound we derived in the previous lemma $\|f_{\Theta_n}(x)\|_{l_1} \leq \eta^d \omega_0^{d-1} |x|$ into (25) we obtain:

$$\begin{aligned} & |(\mathcal{X}_{t+1})_{(i_1, i_2, i_3)} - (\mathcal{X}_t)_{(i_1, i_2, i_3)}| \\ &\leq 2(\eta^d \omega_0^{d-1} + \kappa)|I_1^t + 1 - i_1|(\eta^d \omega_0^{d-1})^2 I_2^t I_3^t \\ &\quad + 2(\eta^d \omega_0^{d-1} + \kappa)|I_2^t + 1 - i_2|(\eta^d \omega_0^{d-1})^2 I_1^t I_3^t \\ &\quad + 2(\eta^d \omega_0^{d-1} + \kappa)|I_3^t + 1 - i_3|(\eta^d \omega_0^{d-1})^2 I_1^t I_2^t \\ &\leq C_2 \max_n (I_n^t + 1 - i_n) \end{aligned} \quad (26)$$

where $C_2 = 6(\eta^{3d} \omega_0^{3d-3} + \kappa \eta^{2d} \omega_0^{2d-2}) \max(I_1^t I_2^t, I_1^t I_3^t, I_2^t I_3^t)$. The proof is completed.

3.5 Continual Learning via Memory Replay

To alleviate forgetting, we design a long-tail memory buffer that utilizes a part of historical data when fitting new data. Theorem 1 shows that more distant information is more likely to be forgotten. Thus we consider the long-tail Beta distribution to construct a memory buffer, which stores more data that is distant from the new data stream. For multi-aspect streams, the memory buffer $\mathcal{M}_t \in \mathbb{R}^{I_1^t \times \dots \times I_N^t}$ at the time t is constructed through a sampling process on the historical data $\mathcal{X}_{t-1}$ by $(\mathcal{M}_t)_{(i_1, \dots, i_N)} = (\mathcal{X}_{t-1})_{(i_1, \dots, i_N)}$ if $(i_1, \dots, i_N) \in \mathcal{I}_t$ otherwise $(\mathcal{M}_t)_{(i_1, \dots, i_N)} = 0$ where

$$\mathcal{I}_t = \{(i_1, \dots, i_N) \mid i_n = \lfloor w_n^j I_n^{t-1} \rfloor, w_n^j \sim \text{Beta}(\alpha, \beta), j = 1, \dots, J, n = 1, \dots, N\}.$$

The $\mathcal{I}_t$ is the index set of the memory buffer such that the indexes in $\mathcal{I}_t$ follow a long-tail Beta distribution to store more information that is distant from the new data stream. This is achieved by setting appropriate α and β such that $\text{Beta}(\alpha, \beta)$ is a long-tail distribution. A total number of J^N indices in $\mathcal{I}_t$ are selected to construct the memory buffer $\mathcal{M}_t$.

The new online optimization model at time t with the memory buffer $\mathcal{M}_t$ is formulated as

$$\min_{\{\Theta_n\}} \|\mathcal{P}_t \otimes \mathcal{Y}_t - [\{f_{\Theta_n}([I_n^t])\}_{n=1}^N]\|_F^2 + \sum_{v \in \mathcal{I}_t} (\mathcal{M}_{t(v)} - [\{f_{\Theta_n}(v(n))\}_{n=1}^N])^2 \quad (27)$$

Given a tensor of size $I_1 \times \dots \times I_N$, OTTD consumes $\mathcal{O}(mrd \sum_{n=1}^N I_n + r^2 \prod_{n=1}^N I_n)$ in each iteration, where $\mathcal{O}(mr^2 d \sum_{n=1}^N I_n)$ is the INR complexity, $\mathcal{O}(r^2 \prod_{n=1}^N I_n)$ is the TT product complexity, r is the TT rank, and m, d are network's width and depth.

4 Experiments

4.1 Experimental Setup

We evaluate the proposed TT-OFTD approach on both single-aspect (e.g., temporal evolution) and multi-aspect (e.g., spatial and temporal evolutions) streaming tensor completion tasks. We use Normalized Reconstruction Error (NRE) as the primary evaluation metric across various sampling rates (e.g., 0.1, 0.2, 0.3).

4.2 Results

[Note to author: Insert your TT-specific experimental results, NRE tables, and temporal curve figures here, directly comparing the TT-OFTD implementation against baseline static tracking algorithms.]

5 Conclusion

We have proposed a novel streaming data completion method, OFTD, which utilizes Tensor-Train (TT) decomposition parameterized by INRs to model tensor streams. By transforming classical online tensor tracking into a continuous functional learning paradigm coupled with long-tail memory replay, the model successfully adapts to new streaming data while mitigating catastrophic forgetting. Future work will explore more advanced, buffer-free continual learning methods such as gradient projection to further optimize the learning of INRs for infinite tensor streams.

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